

Mark Lourenco Rojas

Software Developer
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Expertise

Proficient in:

- Unreal Engine, Unity, Maya 3D
- Substance Painter & Designer
- Visual Studio
- Adobe Photoshop
- Github, Perforce

Familiar with:

- Zbrush
- Marvelous Designer
- Android Studio
- Houdini

Skills

- C++, C#, Unreal Blueprints
- Data Structures and Algorithms, OOP, ECS, Debugging
- AI Behavior, Pathfinding
- Multi-threading, Parallel Programming
- User Interface (UI) Programming
- Augmented Reality (AR), Mobile Development
- Hard-surface modeling, modularization
- Unreal Niagara & Cascade VFX, Unity VFX
- Basic rigging & animation
- Quick prototyping, Game Sense & Design
- Collaboration, Effective Communication with Artists and Programmers

Languages

English - Fluent
Spanish - Fluent

Education

10.2023 / 09.2026

LaSalle College Vancouver

Vancouver, BC
Bachelor's Degree in Computer Games and Programming Skills

05.2026 / 08.2026

Centre for Digital Media

Vancouver, BC
CDM Practicum Program

03.2022 / 03.2023

Vancouver Institute of Media Arts

Vancouver, BC
Diploma in Game Art & Design

Projects

05.2026 / 08.2026

Magic Trails | Educational AR Mobile Experience

Centre for Digital Media Practicum

Gameplay Programmer, Audio Direction/Programmer



- Engineered **AR gameplay mechanics** such as **plane-anchored placement**, **hand-gesture recognition**, and **directional audio beacons**, to ground each mission in the player's real environment.
- Built a **Unity Audio Manager** supporting layered adaptive music with **sample-accurate stem toggling**, spatialized 3D and 2D SFX, and cross-scene music persistence, powering all audio for the app.
- **Led audio design direction** including music, SFX, and dialogue across all missions/menus, alongside programming gameplay systems for a **real external client**.

04.2025 / 06.2026

ML_Engine | Custom C++ / DirectX 11 Game Engine

Independent Development

Engine Programmer, Graphics Programmer



- Built a custom **DirectX 11 rendering pipeline** with hand-written **HLSL shaders** for lighting, shadow mapping, terrain, and post-processing.
- Designed a core engine architecture: **skeletal/code-driven animation**, a **particle system**, and a **data-driven GameObject/Component/Service architecture** with JSON level loading.
- Shipped **five standalone tech demos** over 14 months, each pushing a different system further, from a thermal-vision post-processing shader to a multithreaded physics simulation.

01.2023 / 05.2023

Rute: The Spirit Dish | 3D Stealth-and-Puzzle Adventure

The Rookies

Lead Programmer, Systems Designer, Gameplay Programmer, Gameplay Designer



- **Contributed to game concept, puzzle design, and gameplay flow** during pre-production, **balancing scope against engineering constraints** for a tight timeline.
- Built **player movement/camera control systems** and **stealth mechanics** with enemyAI (detection, tracking, reaction) for third-person exploration.
- Implemented **dialogue and puzzle logic** driving narrative pacing and moment-to-moment challenges.

Hobbies

- Game Jams
- Videogames (RPGs, CRPGs, Auto Chess...)
- TTRPGs (DnD 5e, Alien RPG...)
- Archery
- Music
- Gym

02.2026

Masquerade Murder Night | 2D Side-Scroller Social-Deduction Party Game

Global Game Jam 2026

Gameplay Designer, Level Designer, Gameplay Programmer

- **Collaborated on core game concept, mechanics, and level layout**, establishing the team's design foundation.
- Built core **AI pathfinding** for NPCs using a **node system, with clumping, popular-area gathering, and unpredictable behavior**, so NPCs could pass as real players.
- Implemented **networked gameplay mechanics** to keep behavior **synced across clients**.

12.2022 / 02.2023

Pawl's Impossible Journey | 3D Platform Adventure Game

Vancouver Institute of Media Arts

Gameplay Programmer

- Built **player movement and platforming mechanics** driving third-person traversal through forest environments.
- Implemented **checkpoint, collectible, obstacles, and level-transition systems** structuring the game's progression.
- Built **enemy actors, a damage system, and a dialogue system** driving combat and narrative moments.

09.2025 / 03.2026

The Fool | 2D Deck-Building Roguelite Turn-Based RPG

LaSalle College Vancouver Production

Lead Programmer, Systems Designer, Gameplay Programmer, Gameplay Designer

- Built a **scalable card system** supporting unique icons, pricing, and active/passive effects per card, and **designed a partial card set** with balanced strengths and weaknesses across categories, enabling varied strategy against different enemies and scenarios.
- Developed a **turn-order system** driven by player and enemy speed stats, to govern the flow and pacing of combat.
- Constructed a **scalable enemy system** supporting unique skills and stats per enemy, enabling **varied encounter design without duplicating code**.

